

Data Collection & Preparation

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Team ID	
Project Name	Voice-Based Concept Understanding Analyzer (VBCUA)

Data Collection & Preparation

This section describes how data is sourced, collected, validated, and prepared for use within the Voice-Based Concept Understanding Analyzer (VBCUA), covering both the aspects that are implemented in the current version and those identified as future enhancements.

Data Collection & Preparation	Data source identification (datasets, APIs, user inputs, CSV, etc.): VBCUA does not use a pre-existing dataset, external API, or CSV file. The two data sources are: (1) user-provided audio input — a spoken explanation recorded and uploaded by the student through the Streamlit file uploader, and (2) a locally stored reference answer text file (data/reference_answers/sample_answer.txt) used as the ground-truth standard for comparison. The speech-to-text and semantic similarity models (Whisper “base” and Sentence Transformers all-MiniLM-L6-v2) are pre-trained third-party models used as-is; no custom model training data is collected or used by this project. Data collection method: Audio input is collected at runtime via the Streamlit file uploader (WAV, MP3, or M4A) and saved to the data/audio directory for processing. The reference answer is authored and curated manually as a plain text file rather than collected from an external source. Data cleaning (remove nulls, duplicates, noise): The reference answer loader validates that the file exists and is not empty, raising an error if it is missing or blank (load_reference_answer in file_io.py). The transcript returned by Whisper is stripped of leading and trailing whitespace before further processing. Future enhancement: no explicit audio-level noise reduction or de-duplication is currently implemented. The Librosa library is listed as a dependency for this purpose, but the corresponding module (audio_analysis.py) is an empty placeholder in the current version. Data preprocessing & formatting for model input: Uploaded audio is passed directly to Whisper's transcribe() function, which performs its own internal resampling and feature extraction. The resulting transcript and the reference answer text are validated and stripped of surrounding whitespace (_validate_text in semantic_analyzer.py) before being encoded into embeddings by the Sentence Transformer model for similarity comparison. Data validation and quality check: The file uploader restricts uploads to the supported audio formats (WAV, MP3, M4A) before processing. After transcription, the result is type-checked to confirm it is a string, and a TranscriptionError is raised otherwise. Both the transcript and the reference answer are checked for emptiness before similarity computation, with descriptive errors raised if validation fails. Data privacy & ethical considerations: The current version does not implement authentication, encryption, or an explicit data-retention or deletion policy; uploaded audio files and generated reports persist locally on the host filesystem (data/audio and reports/) until manually removed. The application is intended for controlled, classroom-scale educational use rather than public deployment.
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	Future enhancement: informed consent messaging before recording, configurable data-retention limits, and anonymisation or secure deletion of audio files are identified as necessary additions before any wider or production deployment. Input-output examples: VBCUA does not use prompt-based generation; it performs direct model inference (speech recognition followed by embedding-based similarity), so there is no prompt-engineering step. An illustrative input-output example based on the included sample reference answer: a spoken explanation of inheritance in object-oriented programming is transcribed to text, compared against the stored reference answer describing inheritance, code reuse, and the “is-a” relationship, and returns a similarity score, grade (e.g., “Excellent”), and corresponding feedback text. Dataset documentation: The project currently includes a single reference answer file (sample_answer.txt) covering the concept of inheritance, with no accompanying metadata such as topic tags, author, or last-updated date. Future enhancement: as the number of reference answers grows to cover multiple concepts or subjects, structured documentation (a manifest listing each reference answer's topic, source, and version) is identified as a necessary addition to keep the reference material organised and maintainable.
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